

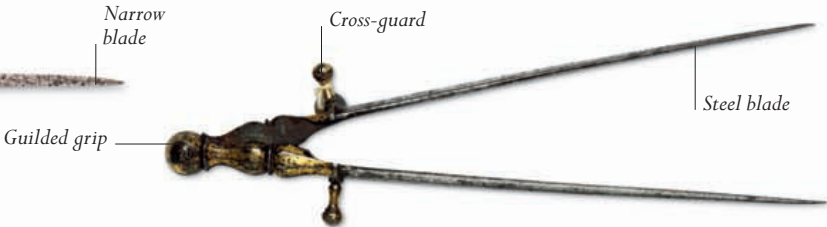
FIELD AND SIEGE ARTILLERY

Through the 17th and 18th centuries, artillerymen continued to use smoothbore cast-bronze and cast-iron firing stone or cast-iron shot. The next breakthrough in artillery design came in the mid-19th century with the development of rifled cannon, improved gunpowder, elongated projectiles, and special instruments. While guns fired in a level trajectory, the lighter, short-barrelled mortar was used to fire in an elevated trajectory – generally more than 45 degrees. Smaller hand mortars were also developed for firing fuzed grenades. These are often named after the Dutch military engineer Menno, baron van Coehoorn, who developed a design for this weapon.

▲ GUNNER'S STILETTO

Date	18th century
Origin	Italy
Weight	150g (5½oz)
Length	34cm (13½in)

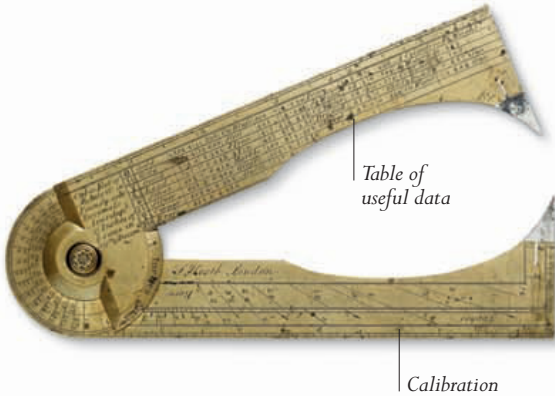
Artillerymen used this specialist dagger for measuring the bore of the gun and the size of the shot. It could also be used for self-defence in close-quarters combat.



► GUNNER'S CALLIPERS

Date	18th century
Origin	England

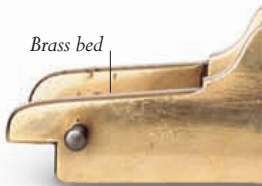
Callipers were used to measure the diameter of the cannonball. When the two arms were crossed over, they could also make internal measurements, such as that of the cannon's bore.



▲ GUNNER'S DIVIDERS

Date	18th century
Origin	England
Length	44cm (17¼in)

Dividers were used with a rule to perform calculations on the proportions of fortifications. This steel pair has a cross-guard, indicating they could also be used as a dagger if the artillery position came under attack.



▲ INDIAN 6-POUNDER

Date	1693–1743
Origin	India
Weight	3.4 tonnes (3.7 tons)
Length	3.86m (12½ft)
Calibre	9.5cm

The bore of this gun is sleeved with an iron tube made from parallel strips welded together. This made the gun and the bore more durable, especially when the gun was used to fire stone shot.



▼ BRONZE THREE-BARRELLED GUN

Date	1704
Origin	France
Weight	905kg (1,996lb)
Length	1.62m (5¼ft)
Calibre	11.5cm

Three barrels, two side by side with the third above, were cast in one piece and could be fired one at a time or simultaneously. The intriguing design did not prove successful in practice, because the gun was difficult to reload and very heavy to manoeuvre.



◀ COEHOORN MORTAR

Date	c.1720
Origin	England
Weight	39kg (86lb)
Length	32cm (12½in)
Calibre	4.5in

The Coehoorn mortar was a small, portable mortar used to despatch grenades. Swiss-born Andrew Schalch, first Master Founder of the Royal Brass Foundry at Woolwich in England, cast this one. It is mounted on its original wooden bed, which is just 30cm (12in) wide and 51cm (20in) long.